



< Previous















Next >

2.2.3 Orthonormal vectors and matrices

 Bookmark this page

Verified Certificate

 THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

 Verified Certificate

Valid Certificate ID

Issued Month / Year

12345678901234567890



Maurilio Pugliesi

Professor

UniversityX



Justine Doe, Ph.D.

Professor

UniversityX

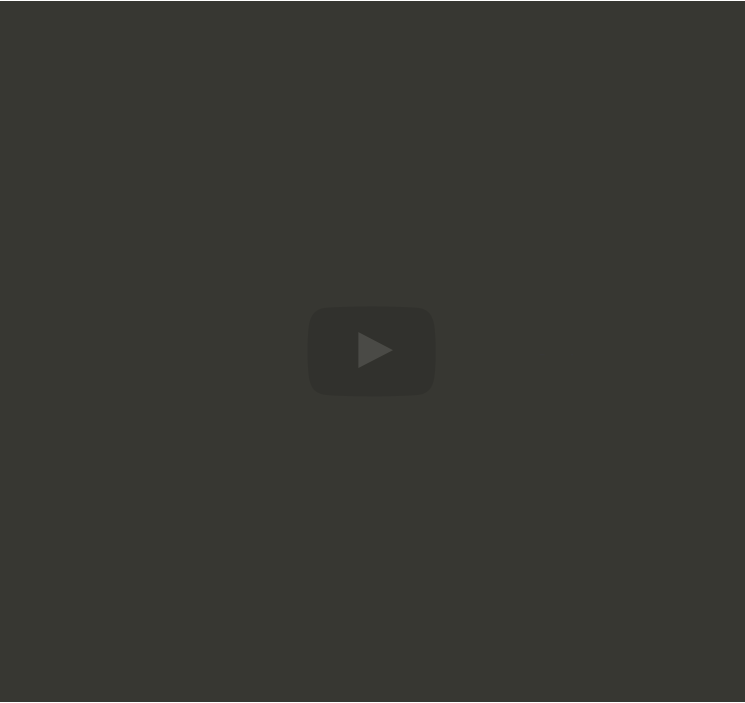
→ Unlimited access to all course materials

→ Shareable [verified certificate](#)

→ AI learning assistant and translations

Upgrade for \$99

Video



▶

5:36 / 5:36

▶ 2.0x

Okay, now for historical reasons when we start talking about such matrices we tend to use the letter q for the columns. So I'm going to change the u here to a q and say okay we have n mutually orthonormal vectors q_0 through q_{n-1} , all in $\mathbb{C}^m$. And we know that they have this property right here. Now let's take those vectors and let's make them columns of a matrix. What happens if you look at Q^H transpose times Q ?

Video

[Download video file](#)

Transcripts

- [Download SubRip \(.srt\) file](#)
- [Download Text \(.txt\) file](#)

Reading assignment

1/1 point (graded)

Read Unit 2.2.3 of the notes. [\[LINK\]](#)

☒

 done

Submit

Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 02 / 02.2.3

Add a Post

Show all posts

by recent activity

There are no posts in this topic yet.

Homework 2.2.3.1

1.0/1.0 point (graded)

Let $u \neq 0 \in \mathbb{C}^m$.

$u / \|u\|_2$ has unit length.

ALWAYS

 Answer: ALWAYS

Solution:

Now prove it!

For complete solution, see [[LINK](#)].

Submit

 Answers are displayed within the problem

Homework 2.2.3.2

1.0/1.0 point (graded)

Let $Q \in \mathbb{C}^{m \times n}$ (with $n \leq m$). Partition $Q = \left(\begin{array}{c|c|c|c} q_0 & q_1 & \cdots & q_{n-1} \end{array} \right)$.

Q is an orthonormal matrix if and only if $q_0, q_1, \dots, q_{n-1}$ are mutually orthonormal.

TRUE

 Answer: TRUE

Solution:

Now prove it!

For complete solution, see [[LINK](#)].

Submit

 Answers are displayed within the problem

Homework 2.2.3.3

1/1 point (graded)

Let $Q \in \mathbb{C}^{m \times n}$ and $Q^H Q = I$.

$Q Q^H = I$.

SOMETIMES

 Answer: SOMETIMES

Solution:

Now explain why!

Now explain why:

For complete solution, see [[LINK](#)].

Submit

 Answers are displayed within the problem

[< Previous](#) [Next >](#)



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)
- [Careers](#)
- [News](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)
- [Sitemap](#)
- [Cookie Policy](#)
- [Your Privacy Choices](#)

Connect

- [Idea Hub](#)
- [Contact Us](#)
- [Help Center](#)
- [Security](#)
- [Media Kit](#)

